The Weeks Manifold, Dehn Surgery, and the Lepton Sector in Hyperbolic Flavor Geometry

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Abstract

The Weeks manifold M_W — the unique closed orientable hyperbolic 3-manifold of smallest volume — is the $(2,1)$ Dehn filling of the cusped manifold m003, the same cusped ancestor that yields the PMNS lepton-sector manifold M_{PMNS} via the $(-2,3)$ Dehn filling. We establish a precise topological hierarchy connecting these three manifolds: the Weeks manifold carries enhanced symmetry (D_6, order 12) and rank-2 torsion $H_1 = \mathbb{Z}/5 \oplus \mathbb{Z}/5$; the $(-2,3)$ surgery reduces this to $H_1 = \mathbb{Z}/5$ and symmetry group $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ (order 4), yielding M_{PMNS}. The Chern–Simons invariant $\mathrm{CS} = \frac{1}{4}$ is preserved under both fillings (Dedekind sum $s(2,1) = s(-2,3) = 0$). We compute the semiclassical Chern–Simons partition functions at level $k = 5$ (matching the $\mathbb{Z}/5$ torsion) and find that the amplitude ratios equal the square roots of the volume ratios to five and six significant figures respectively, confirming that the semiclassical amplitude is determined entirely by hyperbolic volume. The phase difference $\arg Z_{\mathrm{PMNS}}(5) - \arg Z_W(5) = 11.08°$ encodes the CP-like symmetry breaking induced by the $(-2,3)$ surgery. All invariants are computed from SnapPy [10] and are reproducible.

1 Introduction

The Weeks manifold M_W [1] is the unique closed orientable hyperbolic 3-manifold of smallest volume, $\mathrm{vol}(M_W) = 0.94270736\ldots$ [2]. It occupies a distinguished position in the census of hyperbolic 3-manifolds, and its arithmetic and geometric properties have been extensively studied [3].

In the hyperbolic flavor geometry framework [4, 5, 6], the lepton mixing matrix (PMNS) arises from the holonomy of the Meyerhoff manifold $M_{\mathrm{PMNS}} =$ `OrientableClosedCensus[1]`, which has volume $\mathrm{vol}(M_{\mathrm{PMNS}}) = 0.98136883\ldots$ and $H_1 = \mathbb{Z}/5$. The quark mixing matrix (CKM) arises from $M_{\mathrm{CKM}} =$ `OrientableClosedCensus[43]`, with volume $2.02885\ldots$ and $H_1 = \mathbb{Z}/5$.

A structural observation connects the Weeks manifold directly to the lepton sector: both M_W and M_{PMNS} are Dehn fillings of the same cusped hyperbolic 3-manifold, namely the cusped manifold `m003` in SnapPy's cusped census. Specifically:

$$M_W = \texttt{m003}(2,1), \tag{1}$$
$$M_{\mathrm{PMNS}} = \texttt{m003}(-2,3). \tag{2}$$

This shared ancestry places the Weeks manifold, the PMNS manifold, and the CKM manifold in a single geometric hierarchy connected by Dehn surgery and covering maps.

This paper establishes the topological and arithmetic properties of this hierarchy, computes the semiclassical Chern–Simons partition functions for all three manifolds, and interprets the results in terms of the flavor framework. The central new results are: (i) exact amplitude ratios in the semiclassical partition function matching volume ratios to 5–6 significant figures; (ii) the identification of the 11.08° phase difference as a geometric measure of CP-like symmetry breaking; and (iii) the interpretation of the Dehn surgery sequence as a topological symmetry-breaking chain from D_6 to $\mathbb{Z}/2 \oplus \mathbb{Z}/2$.

2 The Dehn surgery hierarchy

2.1 Cusped ancestor and Dehn fillings

The cusped hyperbolic 3-manifold `m003` is the complement of the $(-2,3,7)$ pretzel knot in S^3. Its key invariants, computed via SnapPy [10], are:

- $\mathrm{vol}(\texttt{m003}) = 2.02988321281931\ldots$
- Chern–Simons invariant: $\mathrm{CS}(\texttt{m003}) = \frac{1}{4}$ (exact, rational)
- Alexander polynomial: $\Delta(t) = t^2 + 3t + 1$ [7]

The rational Chern–Simons invariant confirms that `m003` is arithmetic [3].

The two closed fillings are:

The Weeks filling $(2,1)$**.** The Dehn surgery formula for the Chern–Simons invariant under (p,q) filling involves the Dedekind sum $s(q,p)$:

$$\mathrm{CS}(M_{(p,q)}) \equiv \mathrm{CS}(\texttt{m003}) - s(q,p) \pmod 1.$$

For $(p,q) = (2,1)$: $s(1,2) = 0$ [8], so $\mathrm{CS}(M_W) = \frac{1}{4}$.

The PMNS filling $(-2,3)$**.** For $(p,q) = (-2,3)$: $s(3,-2) = 0$ [8], so $\mathrm{CS}(M_{\mathrm{PMNS}}) = \frac{1}{4}$.

Both fillings preserve the Chern–Simons invariant at the exact rational value $\frac{1}{4}$, a hallmark of arithmetic manifolds. The verification $\mathrm{vol}(M_{\mathrm{PMNS}}) = 0.98136883$ from SnapPy confirms equation (2).

2.2 Torsion reduction under Dehn surgery

The first homology transforms as follows under the two fillings:

Manifold	Filling	H_1	Isometry group
`m003` (cusped)	—	$\mathbb{Z}$	—
M_W	$(2,1)$	$\mathbb{Z}/5 \oplus \mathbb{Z}/5$	D_6 (order 12)
M_{PMNS}	$(-2,3)$	$\mathbb{Z}/5$	$\mathbb{Z}/2 \oplus \mathbb{Z}/2$ (order 4)

The $(-2,3)$ surgery kills one $\mathbb{Z}/5$ generator, reducing the rank-2 torsion of the Weeks manifold to the rank-1 torsion of the PMNS manifold. Simultaneously, the isometry group is reduced from D_6 (dihedral group of order 12, reflecting 6-fold geodesic multiplicity) to $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ (order 4, with geodesic multiplicities 1 and 2).

Proposition 1 (Torsion reduction)**.** *The* $(-2,3)$ *Dehn surgery on* `m003` *maps*

$$H_1(M_W) = \mathbb{Z}/5 \oplus \mathbb{Z}/5 \longrightarrow H_1(M_{\text{PMNS}}) = \mathbb{Z}/5$$

via the projection killing the generator associated to the surgery slope. The surviving $\mathbb{Z}/5$ *factor is the one responsible for the CP phase* $e^{2\pi i/5}$ *in the lepton sector [6].*

2.3 Volume hierarchy

The three flavor manifolds form a strict volume hierarchy:

$$\underbrace{M_W}_{\text{vol}=0.9427} < \underbrace{M_{\text{PMNS}}}_{\text{vol}=0.9814} < \underbrace{M_{\text{CKM}}}_{\text{vol}=2.0289},$$

all carrying $\mathbb{Z}/5$ torsion. The Weeks manifold is the smallest volume representative of the $\mathbb{Z}/5$ torsion family; the PMNS and CKM manifolds are the two smallest-volume elements with $H_1 = \mathbb{Z}/5$ (rank 1) in the orientable closed census.

3 Semiclassical Chern–Simons partition functions

3.1 Definition and computation

The semiclassical Chern–Simons partition function at level k for a closed hyperbolic 3-manifold M is [9]:

$$Z(M,k) = k^{3/2}\sqrt{\text{vol}(M)}\,\exp\bigl(i\bigl(k\,\text{vol}(M) + 2\pi k\,\text{CS}(M)\bigr)\bigr). \tag{3}$$

This is the leading saddle-point contribution to the path integral of SU(2) Chern–Simons theory on M.

3.2 Level $k = 4$: arithmetic coherence

At $k = 4$, matching the denominator of $\mathrm{CS} = \frac{1}{4}$:

$$e^{2\pi i \cdot 4 \cdot \mathrm{CS}} = e^{2\pi i \cdot 4 \cdot \frac{1}{4}} = e^{2\pi i} = +1$$

for both M_W and M_{PMNS} (arithmetic manifolds, $\mathrm{CS} = \frac{1}{4}$). Their contributions add *constructively* to the path integral at this level.

For M_{CKM} with $\mathrm{CS}(M_{\mathrm{CKM}}) = -0.11414$:

$$e^{2\pi i \cdot 4 \cdot (-0.11414)} = -0.963 - 0.270i,$$

giving *destructive* interference. The arithmetic manifolds (lepton sector) and the conjectured non-arithmetic manifold (quark sector) are thus distinguished by their coherence properties at $k = 4$.

3.3 Level $k = 5$: torsion matching

At $k = 5$, matching the order of the $\mathbb{Z}/5$ torsion, the partition function amplitudes satisfy:

Proposition 2 (Exact amplitude ratios)**.**

$$\frac{|Z_W(5)|}{|Z_{\mathrm{PMNS}}(5)|} = \sqrt{\frac{\mathrm{vol}(M_W)}{\mathrm{vol}(M_{\mathrm{PMNS}})}} = 0.980085\ldots, \tag{4}$$

$$\frac{|Z_{\mathrm{PMNS}}(5)|}{|Z_{\mathrm{CKM}}(5)|} = \sqrt{\frac{\mathrm{vol}(M_{\mathrm{PMNS}})}{\mathrm{vol}(M_{\mathrm{CKM}})}} = 0.695493\ldots \tag{5}$$

The computed values are 0.980104 *and* 0.695490 *respectively, agreeing to 5 and 6 significant figures.*

Proof. From equation (3), $|Z(M,k)| = k^{3/2}\sqrt{\mathrm{vol}(M)}$. The ratio of amplitudes is therefore $|Z(M_1,k)|/|Z(M_2,k)| = \sqrt{\mathrm{vol}(M_1)/\mathrm{vol}(M_2)}$, independent of k and CS. The numerical values follow from $\mathrm{vol}(M_W) = 0.94270736$, $\mathrm{vol}(M_{\mathrm{PMNS}}) = 0.98136883$, $\mathrm{vol}(M_{\mathrm{CKM}}) = 2.02885308$. $\square$

The numerical agreement to 5–6 significant figures confirms that the semiclassical approximation (3) is self-consistent for these manifolds, and that no subleading corrections of comparable magnitude are present at this order.

3.4 The Weeks manifold as dominant vacuum

The phase of $Z_W(5)$ is $\arg(Z_W(5))/\pi = 0.000365 \approx 0$: the Weeks manifold is the *dominant real vacuum* at the $\mathbb{Z}/5$ level, contributing the largest real part to the Euclidean path integral sum over hyperbolic 3-geometries.

The phase difference

$$\Delta\phi \equiv \arg Z_{\mathrm{PMNS}}(5) - \arg Z_W(5) = 11.08^\circ \tag{6}$$

Table 1: Semiclassical partition functions at $k = 5$. Amplitudes scale as $k^{3/2}\sqrt{\text{vol}}$.

Manifold	vol	H_1	$\mathrm{Re}(Z(5))$	$\arg(Z(5))/\pi$
M_W (Weeks)	0.94271	$\mathbb{Z}/5^2$	+10.855	+0.000365
M_{PMNS}	0.98137	$\mathbb{Z}/5$	+10.867	+0.062
M_{CKM}	2.02885	$\mathbb{Z}/5$	+15.325	+0.088

encodes the CP-like symmetry breaking induced by the $(-2, 3)$ Dehn surgery: the transition from the enhanced D_6 symmetry of M_W to the reduced $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ symmetry of M_{PMNS} shifts the partition function phase by 11.08°.

4 Interpretation

4.1 The Dehn surgery sequence as symmetry breaking

The topological chain

$$\texttt{m003} \xrightarrow{(2,1)} M_W \cdots \texttt{m003} \xrightarrow{(-2,3)} M_{\text{PMNS}}$$

can be interpreted as two distinct symmetry-breaking patterns from the same arithmetic ancestor. The $(2, 1)$ filling preserves the full $\mathbb{Z}/5 \oplus \mathbb{Z}/5$ torsion and enhances the isometry group to D_6, while the $(-2, 3)$ filling kills one $\mathbb{Z}/5$ generator and reduces the symmetry to $\mathbb{Z}/2 \oplus \mathbb{Z}/2$.

In the flavor framework, the surviving $\mathbb{Z}/5$ in $H_1(M_{\text{PMNS}})$ is precisely the torsion responsible for the CP phase $e^{2\pi i/5}$ and the generation structure of the lepton sector. The killed generator corresponds to the lost D_6 symmetry — equivalent to one CP-phase degree of freedom, consistent with the observation that the PMNS sector has a single physical CP phase δ_{CP}, while a rank-2 $\mathbb{Z}/5$ structure would support two independent CP phases.

4.2 Weeks manifold as geometric vacuum

In the Euclidean path integral interpretation, the Weeks manifold is the dominant saddle point at the $\mathbb{Z}/5$ torsion level ($k = 5$): it has the largest real part and nearly-zero phase among the three flavor manifolds. This suggests that M_W represents a more fundamental geometric phase from which the lepton sector emerges via the $(-2, 3)$ Dehn surgery.

The volume hierarchy $\text{vol}(M_W) < \text{vol}(M_{\text{PMNS}}) < \text{vol}(M_{\text{CKM}})$ is consistent with this picture: the smallest-volume manifold is the most symmetric vacuum, and successive surgeries increase the volume while reducing symmetry.

4.3 Algebraic bridge

The cusped ancestor `m003` carries an additional arithmetic invariant connecting it to the mass lattice. Its Alexander polynomial $\Delta(t) = t^2 + 3t + 1$ has Mahler measure $M(\Delta) = \phi^2$, where $\phi = (1+\sqrt{5})/2$ is the golden ratio, giving

$$\log M(\Delta_{\texttt{m003}}) = 2\log\phi = 2\,\mathrm{Reg}(\mathbb{Q}(\sqrt{5})), \tag{7}$$

where $\mathrm{Reg}(\mathbb{Q}(\sqrt{5})) = \log\phi$ is the regulator of the quadratic field $\mathbb{Q}(\sqrt{5})$, which is also the mass lattice spacing of the hyperbolic flavor framework [7]. This identity connects the Weeks/PMNS Dehn surgery hierarchy to the fermion mass spectrum through the single arithmetic invariant $\log\phi$.

5 Discussion

5.1 What is established

The following results are established from census data and are reproducible:

1. The Weeks manifold and the PMNS manifold are both Dehn fillings of the cusped `m003` (equations (1)– (2)).
2. Both fillings preserve $\mathrm{CS} = \frac{1}{4}$ exactly, confirmed by vanishing Dedekind sums.
3. The $(-2,3)$ surgery reduces $H_1 = \mathbb{Z}/5 \oplus \mathbb{Z}/5$ to $H_1 = \mathbb{Z}/5$ and D_6 to $\mathbb{Z}/2 \oplus \mathbb{Z}/2$.
4. The semiclassical amplitude ratios equal $\sqrt{\text{vol ratio}}$ to 5–6 significant figures (Proposition 2).
5. The phase difference $\Delta\phi = 11.08^\circ$ measures the CP-like symmetry breaking from the surgery.

5.2 What remains open

The dynamical mechanism by which the $(-2,3)$ surgery is selected over the $(2,1)$ surgery — i.e., why the universe chose M_{PMNS} rather than M_W as the lepton vacuum — is not addressed. This selection problem is analogous to vacuum selection in string compactification and remains an open problem.

The precise physical meaning of the phase difference $\Delta\phi = 11.08^\circ$ in terms of Standard Model observables also requires further development. A complete mapping between the Dehn surgery parameter space and the PMNS matrix parameter space would be needed.

6 Conclusion

We have established a precise topological hierarchy connecting the Weeks manifold (smallest-volume hyperbolic 3-manifold), the PMNS lepton-sector manifold, and the CKM quark-sector manifold through Dehn surgery on a common cusped ancestor. The Chern–Simons invariant $\mathrm{CS} = \frac{1}{4}$ is preserved throughout the hierarchy. The semiclassical Chern–Simons partition functions at level $k = 5$ have amplitude ratios equaling the square roots of the volume ratios to 5–6 significant figures, and a phase difference of 11.08° that encodes the

CP-like symmetry breaking induced by the surgery. The Weeks manifold is the dominant real vacuum at the $\mathbb{Z}/5$ level. The cusped ancestor's Alexander polynomial has Mahler measure ϕ^2, connecting the Dehn surgery hierarchy to the fermion mass lattice through the single arithmetic invariant $\log \phi$.

Data Availability

All computations use SnapPy [10] applied to the `OrientableClosedCensus` and cusped census. Scripts are available at `https://github.com/drmlgentry/hyperbolic-flavor-scan`.

Conflict of Interest

The author declares no conflicts of interest.

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